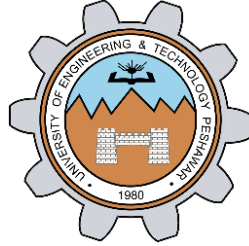


Tracing DNS with Wireshark

LAB # 05



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CSE-303L Data Communication & Networks Lab

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“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

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The Domain Name System (DNS) translates hostnames to IP addresses, fulfilling a critical role in the Internet infrastructure. In this lab, we'll take a closer look at the client side of DNS. Recall that the client's role in the DNS is relatively simple – a client sends a query to its local DNS server, and receives a response back.

The hierarchical DNS servers communicate with each other to either recursively or iteratively resolve the client's DNS query. From the DNS client's standpoint, however, the protocol is quite simple – a query is formulated to the local DNS server and a response is received from that server.

Tracing DNS with Wireshark

- Open Wireshark and enter "ip.addr == your_IP_address" into the filter, where you obtain your_IP_address with ipconfig. This filter removes all packets that neither originate nor are destined to your host.
- Start packet capture in Wireshark.
- With your browser, visit the Web page: <http://www.ietf.org>
- Stop packet capture.

To print a packet, use File->Print, choose Selected packet only, choose Packet summary line, and select the minimum amount of packet detail that you need to answer the question.

1. Locate the DNS query and response messages. Are then sent over UDP or TCP?

Ans: UDP

6863...	587.039992	192.168.26.216	192.168.26.210	DNS	72 Standard query	0xa078 AAAA www.ietf.org
6863...	587.040311	192.168.26.216	192.168.26.210	DNS	72 Standard query	0xdaeb HTTPS www.ietf.org
6865...	587.285878	192.168.26.210	192.168.26.216	DNS	128 Standard query response	0x7c15 AAAA www.ietf.org
6872...	588.089130	192.168.26.216	192.168.26.210	DNS	74 Standard query	0xdfcd A login.live.com
6872...	588.089497	192.168.26.216	192.168.26.210	DNS	74 Standard query	0xb860 AAAA login.live.com
6872...	588.190979	192.168.26.216	192.168.26.210	DNS	74 Standard query	0xdfcd A login.live.com
6872...	588.191106	192.168.26.216	192.168.26.210	DNS	74 Standard query	0xb860 AAAA login.live.com
6873...	588.268820	192.168.26.216	192.168.26.210	DNS	85 Standard query	0x7240 A login.microsoftonline.com
6873...	588.273794	192.168.26.210	192.168.26.216	DNS	345 Standard query response	0x7240 A login.microsoftonline.com
▶ Frame 686312: 72 bytes on wire (576 bits), 72 bytes captured (576 bits) on interface \Device\NPF{...}						0000 7a b3 56 d0 a3 ad cc b0 da ba
▶ Ethernet II, Src: LiteonTechno_ba:0b:bd (cc:b0:da:ba:0b:bd), Dst: 7a:b3:56:d0:a3:ad (7						0010 00 3a 70 ad 00 00 80 11 13 0b
▶ Internet Protocol Version 4, Src: 192.168.26.216, Dst: 192.168.26.210						0020 1a d2 ee 30 00 35 00 26 97 68
▶ User Datagram Protocol, Src Port: 60976, Dst Port: 53						0030 00 00 00 00 00 03 77 77 77
Source Port: 60976						0040 6f 72 67 00 00 1c 00 01
Destination Port: 53						
Length: 38						
Checksum: 0x9768 [unverified]						
[Checksum Status: Unverified]						
[Stream index: 331]						
▶ [Timestamps]						
UDP payload (30 bytes)						
▶ Domain Name System (query)						

- What is the destination port for the DNS query message? What is the source port of DNS response message?

Ans: Source Port is **53** and Destination Port is **53**

```

User Datagram Protocol, Src Port: 60976, Dst Port: 53
  Source Port: 60976
  Destination Port: 53
  Length: 38
  Checksum: 0x9768 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 331]
  ▶ [Timestamps]
  UDP payload (30 bytes)
  Domain Name System (query)

```

- To what IP address is the DNS query message sent? Use ipconfig to determine the IP address of your local DNS server. Are these two IP addresses the same?

Ans: The DNS query message is sent to 192.168.26.210. It is the same IP address as my local DNS server.

No.	Time	Source	Destination	Protocol	Length	Info
6814...	581.656350	192.168.26.216	192.168.26.210	DNS	72	Standard query 0x94b5 HTTPS www.bing.com
6831...	583.485090	192.168.26.210	192.168.26.216	DNS	329	Standard query response 0x5b96 AAAA www.bing.com
6831...	583.485090	192.168.26.210	192.168.26.216	DNS	333	Standard query response 0xb7ab A www.bing.com
6843...	584.687874	192.168.26.216	192.168.26.210	DNS	72	Standard query 0x7c15 AAAA www.ietf.org
6843...	584.688037	192.168.26.216	192.168.26.210	DNS	72	Standard query 0xe489 A www.ietf.org
6843...	584.688180	192.168.26.216	192.168.26.210	DNS	72	Standard query 0x8e59 HTTPS www.ietf.org
6843...	584.694820	192.168.26.210	192.168.26.216	DNS	104	Standard query response 0xe489 A www.ietf.org
6863...	587.039992	192.168.26.216	192.168.26.210	DNS	72	Standard query 0xa078 AAAA www.ietf.org

```

DHCP Server . . . . . : 192.168.26.210
DHCPv6 IAID . . . . . : 449622234
DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-A6-77-4A-CC-B0-DA-BA-0B-BD
DNS Servers . . . . . : 192.168.26.210
NetBIOS over Tcpi. . . . . : Enabled

```

- Examine the DNS query message. What “Type” of DNS query is it? Does the query message contains any “answers”?

Ans: The type of DNS query is ‘A’ and DNS query message contains no answers.

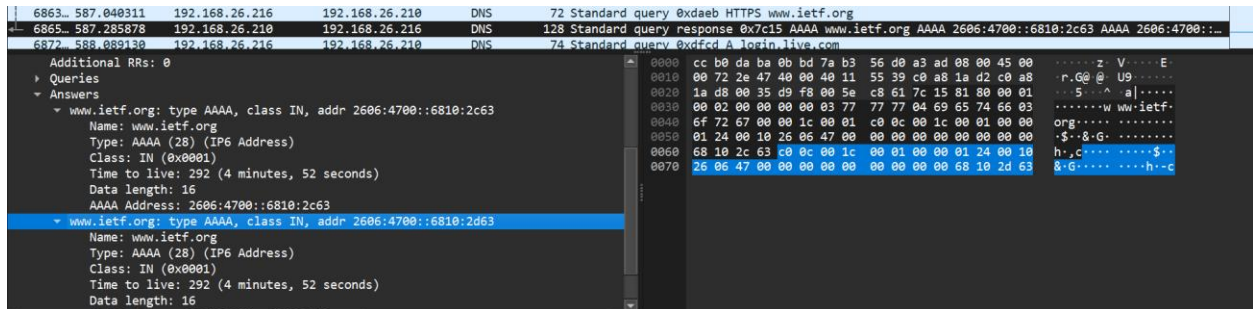
```

Domain Name System (query)
  Transaction ID: 0xa078
  ▶ Flags: 0x0100 Standard query
  Questions: 1
  Answer RRs: 0
  Authority RRs: 0
  Additional RRs: 0
  ▶ Queries
    ▶ www.ietf.org: type AAAA, class IN
    [Response In: 689035]

```

5. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain?

Ans: Two Answers are provided.



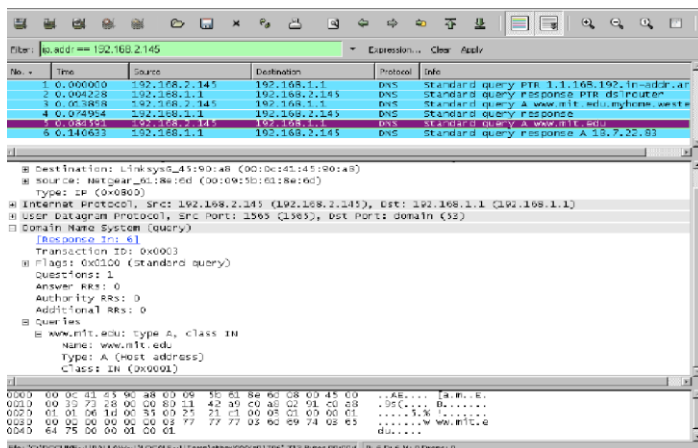
Each answer in the “Answers” section is represented by an RR. An RR consists of several fields, including:

- **Name:** The domain name for which the query was made (e.g., “www.example.com”).
- **Type:** The type of DNS record associated with the domain (e.g., A, AAAA, CNAME, MX).
- **Class:** The class of the record (usually “IN” for Internet).
- **TTL (Time to Live):** How long the answer can be cached (in seconds).
- **Data Length:** The length of the data field.
- **Data:** The actual data associated with the answer (e.g., IP address, domain name).

Now let’s play with *nslookup*.

- Start packet capture.
- Do an *nslookup* on www.mit.edu
- Stop packet capture.

You should get a trace that looks something like the following:



We see from the above screenshot that *nslookup* actually sent three DNS queries and received three DNS responses. For the purpose of this assignment, in answering the following questions, ignore the first two sets of queries/responses, as they are specific to *nslookup* and are not normally generated by standard Internet applications. You should instead focus on the last query and response messages.

1. What is the destination port for the DNS query message? What is the source port of DNS response message?

Ans: Destination port of query: 53

Source port of response: 53

```

4542 74.183330 192.168.26.216 192.168.26.210 DNS 71 Standard query 0x0002 A www.mit.edu
4544 74.696153 192.168.26.210 192.168.26.216 DNS 160 Standard query response 0x0002 A www.mit.edu CNAME www.mit.edu.edgekey.net CNAME e9566.dscb...
4545 74.705392 192.168.26.216 192.168.26.210 DNS 71 Standard query 0x0003 AAAA www.mit.edu
4546 74.867370 192.168.26.210 192.168.26.216 DNS 200 Standard query response 0x0003 AAAA www.mit.edu
5178 89.780010 192.168.26.216 192.168.26.210 DNS 72 Standard query 0x99b5 AAAA l.clarity.ms
5179 89.780186 192.168.26.216 192.168.26.210 DNS 72 Standard query 0x7507 A l.clarity.ms
5180 89.780326 192.168.26.216 192.168.26.210 DNS 72 Standard query 0x82c9 HTTPS l.clarity.ms
5193 89.895983 192.168.26.210 192.168.26.216 DNS 208 Standard query response 0x99b5 AAAA l.clarity.ms CNAME vmss-clarity-ingest-eus-e.eastus.clouda...
5194 89.907731 192.168.26.210 192.168.26.216 DNS 153 Standard query response 0x7507 A l.clarity.ms CNAME vmss-clarity-ingest-eus-e.eastus.clouda...
5230 90.948277 192.168.26.216 192.168.26.210 DNS 80 Standard query 0xbdf3 AAAA beacons.gcp.gvt2.com
5231 90.948835 192.168.26.216 192.168.26.210 DNS 80 Standard query 0x5557 A beacons.gcp.gvt2.com
5232 90.949509 192.168.26.216 192.168.26.210 DNS 80 Standard query 0x358b HTTPS beacons.gcp.gvt2.com
5234 91.011131 192.168.26.210 192.168.26.216 DNS 138 Standard query response 0xbdf3 AAAA beacons.gcp.gvt2.com
5235 91.016767 192.168.26.210 192.168.26.216 DNS 126 Standard query response 0x5557 A beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com A...

> Frame 4542: 71 bytes on wire (568 bits), 71 bytes captured (568 bits) on interface \Device\NPF{...}
> Ethernet II, Src: LiteonTechno_ba:0b:bd (cc:b0:da:ba:0b:bd), Dst: 7a:b3:56:d0:a3:ad (78:00:00:00:00:00)
> Internet Protocol Version 4, Src: 192.168.26.216, Dst: 192.168.26.210
> User Datagram Protocol, Src Port: 54779, Dst Port: 53
  Source Port: 54779
  Destination Port: 53
  Length: 37
  Checksum: 0x1fcb [unverified]
  [Checksum Status: Unverified]
  [Stream index: 109]
  [Timestamps]
  UDP payload (29 bytes)
  Domain Name System (query)

```

```

4544 74.696153 192.168.26.210 192.168.26.216 DNS 160 Standard query response 0x0002 A www.mit.edu CNAME www.mit.edu.edgekey.net CNAME e9566.dscb...
4545 74.705392 192.168.26.216 192.168.26.210 DNS 71 Standard query 0x0003 AAAA www.mit.edu
4546 74.867370 192.168.26.210 192.168.26.216 DNS 200 Standard query response 0x0003 AAAA www.mit.edu CNAME www.mit.edu.edgekey.net CNAME e9566.d...
5178 89.780010 192.168.26.216 192.168.26.210 DNS 72 Standard query 0x99b5 AAAA l.clarity.ms
5179 89.780186 192.168.26.216 192.168.26.210 DNS 72 Standard query 0x7507 A l.clarity.ms
5180 89.780326 192.168.26.216 192.168.26.210 DNS 72 Standard query 0x82c9 HTTPS l.clarity.ms
5193 89.895983 192.168.26.210 192.168.26.216 DNS 208 Standard query response 0x99b5 AAAA l.clarity.ms CNAME vmss-clarity-ingest-eus-e.eastus.clouda...
5194 89.907731 192.168.26.210 192.168.26.216 DNS 153 Standard query response 0x7507 A l.clarity.ms CNAME vmss-clarity-ingest-eus-e.eastus.clouda...
5230 90.948277 192.168.26.216 192.168.26.210 DNS 80 Standard query 0xbdf3 AAAA beacons.gcp.gvt2.com
5231 90.948835 192.168.26.216 192.168.26.210 DNS 80 Standard query 0x5557 A beacons.gcp.gvt2.com
5232 90.949509 192.168.26.216 192.168.26.210 DNS 80 Standard query 0x358b HTTPS beacons.gcp.gvt2.com
5234 91.011131 192.168.26.210 192.168.26.216 DNS 138 Standard query response 0xbdf3 AAAA beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com A...
5235 91.016767 192.168.26.210 192.168.26.216 DNS 126 Standard query response 0x5557 A beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com A...

> Frame 4544: 160 bytes on wire (1280 bits), 160 bytes captured (1280 bits) on interface \Device\NPF{...}
> Ethernet II, Src: 7a:b3:56:d0:a3:ad (7a:b3:56:d0:a3:ad), Dst: LiteonTechno_ba:0b:bd (cc:b0:da:ba:0b:bd)
> Internet Protocol Version 4, Src: 192.168.26.210, Dst: 192.168.26.216
> User Datagram Protocol, Src Port: 53, Dst Port: 54779
  Source Port: 53
  Destination Port: 54779
  Length: 126
  Checksum: 0xa8db [unverified]
  [Checksum Status: Unverified]
  [Stream index: 109]
  [Timestamps]
  UDP payload (118 bytes)
  Domain Name System (response)

```

2. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server?

Ans: Yes, it is same as default local DNS server(192.168..26.210).

No.	Time	Source	Destination	Protocol	Length	Info
3861	49.478179	2401:ba80:aa00:9a02...	2603:1046:1400:1::1	TCP	109	[TCP Retran
3862	49.481176	2603:1046:1400:1::1	2401:ba80:aa00:9a02...	TCP	74	443 → 58745
3863	49.567774	2600:1413:b000:1d::...	2401:ba80:aa00:9a02...	TCP	105	[TCP Retran
3864	49.697475	2603:1046:1400:1::1	2401:ba80:aa00:9a02...	TCP	86	[TCP Dup AC
3865	52.603772	192.168.26.216	23.213.43.218	TCP	54	58779 → 443
3866	52.618868	192.168.26.216	192.168.26.210	DNS	87	Standard qu
3867	52.665258	192.168.26.210	192.168.26.216	DNS	164	Standard qu
3868	52.670921	192.168.26.216	192.168.26.210	DNS	71	Standard qu
3869	52.762706	23.213.43.218	192.168.26.216	TCP	54	443 → 58779
3870	52.762777	192.168.26.216	23.213.43.218	TCP	54	[TCP Dup AC
3871	52.762985	23.213.43.218	192.168.26.216	TCP	85	[TCP Out-Of
3872	52.763040	192.168.26.216	23.213.43.218	TCP	54	58779 → 443
3873	54.678324	192.168.26.216	192.168.26.210	DNS	71	Standard qu
3874	54.684667	192.168.26.210	192.168.26.216	DNS	160	Standard qu
3875	54.823158	192.168.26.210	192.168.26.216	DNS	200	Standard qu

▶ Frame 3873: 71 bytes on wire (568 bits), 71 bytes captured (568 bits) on interface \De
 ▶ Ethernet II, Src: LiteonTechno_ba:0b:bd (cc:b0:da:ba:0b:bd), Dst: 7a:b3:56:d0:a3:ad (7
 ▶ Internet Protocol Version 4, Src: 192.168.26.216, Dst: 192.168.26.210
 ▶ User Datagram Protocol, Src Port: 52336, Dst Port: 53
 ▶ Domain Name System (query)

```

DHCP Server . . . . . : 192.168.26.210
DHCPv6 IAID . . . . . : 449622234
DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-A6-77-4A-CC-B0-DA-BA-0B-BD
DNS Servers . . . . . : 192.168.26.210
NetBIOS over Tcpi. . . . . : Enabled

```

- Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?

Ans: The type of DNS query is ‘A’ and DNS query message contains no answers.

3872	52.763040	192.168.26.216	23.213.43.218	TCP	54	58779 → 443 [RST, ACK] Seq=2 Ack=32 Win=0 Len=0
3873	54.678324	192.168.26.216	192.168.26.210	DNS	71	Standard query 0x0003 AAAA www.mit.edu
3874	54.684667	192.168.26.210	192.168.26.216	DNS	160	Standard query response 0x0002 A www.mit.edu CNAME www.mit.e
3875	54.823158	192.168.26.210	192.168.26.216	DNS	200	Standard query response 0x0003 AAAA www.mit.edu CNAME www.mi

▶ User Datagram Protocol, Src Port: 52336, Dst Port: 53
 ▶ Domain Name System (query)
 Transaction ID: 0x0003
 Flags: 0x0100 Standard query
 Questions: 1
 Answer RRs: 0
 Authority RRs: 0
 Additional RRs: 0
 ▶ Queries
 www.mit.edu: type AAAA, class IN
 Name: www.mit.edu
 [Name Length: 11]
 [Label Count: 3]
 Type: AAAA (28) (IPv6 Address)
 Class: IN (0x0001)
 [Response In: 3875]

0000 7a b3 56 d0 a3 ad cc b0 da ba 0b bd 08 00
 0010 00 39 82 f1 00 00 80 11 00 c8 c0 a8 1a d8
 0020 1a d2 cc 70 00 35 00 25 0e 55 00 03 01 00
 0030 00 00 00 00 00 03 77 77 77 03 6d 69 74
 0040 64 75 00 00 1c 00 01

4. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain?

Ans: It contains 3 answers.

```
3870 52.762777 192.168.26.216 23.213.43.218 TCP 54 [TCP Dup ACK 3865#1] 58779 → 443 [ACK] Seq=2 Ack=1 Win=1024 Len=0
3871 52.762985 23.213.43.218 192.168.26.216 TCP 85 [TCP Out-Of-Order] 443 → 58779 [PSH, ACK] Seq=1 Ack=2 Win=501 Len=31
3872 52.763040 192.168.26.216 23.213.43.218 TCP 54 58779 → 443 [RST, ACK] Seq=2 Ack=32 Win=0 Len=0
3873 54.678324 192.168.26.216 192.168.26.210 DNS 71 Standard query 0x0003 AAAA www.mit.edu
3874 54.684667 192.168.26.210 192.168.26.216 DNS 160 Standard query response 0x0002 A www.mit.edu CNAME www.mit.edu.edgekey
3875 54.823158 192.168.26.210 192.168.26.216 DNS 200 Standard query response 0x0003 AAAA www.mit.edu CNAME www.mit.edu.edgekey

Answer RRs: 3
Authority RRs: 0
Additional RRs: 0
Queries
  www.mit.edu: type A, class IN
    Name: www.mit.edu
    [Name Length: 11]
    [Label Count: 3]
    Type: A (1) (Host Address)
    Class: IN (0x0001)
Answers
  www.mit.edu: type CNAME, class IN, cname www.mit.edu.edgekey.net
  www.mit.edu.edgekey.net: type CNAME, class IN, cname e9566.dscb.akamaiedge.net
  e9566.dscb.akamaiedge.net: type A, class IN, addr 104.85.18.51
[Request In: 3868]
[Time: 2.013746000 seconds]
```

Each answer in the “Answers” section is represented by an RR. An RR consists of several fields, including:

- **Name:** The domain name for which the query was made (e.g., “www.example.com”).
- **Type:** The type of DNS record associated with the domain (e.g., A, AAAA, CNAME, MX).
- **Class:** The class of the record (usually “IN” for Internet).
- **TTL (Time to Live):** How long the answer can be cached (in seconds).
- **Data Length:** The length of the data field.
- **Data:** The actual data associated with the answer (e.g., IP address, domain name).